

CS516: Parallelization of Programs

End Sem Examination (2024-25-W Semester)

Max. Points: 100

Duration: 2.5 hours

May 1, 2025

Instructions:

1. Make your assumptions and state them wherever necessary.
2. Don't ask unnecessary doubts during the exam. You should contact the instructor only in case of possible correction, ambiguity, or missing information in the paper.
3. All the questions are compulsory.

Question-1:

[25 Points]

Consider the following loop nest.

```
for (int i=-10; i<100; i++) {  
    for (int j=-10; j<100; j++) {  
        for (k=-10; k<100; k++) {  
S1:            A[i-2][j-2][k+3] = D[i-3][j+1][k-2];  
S2:            C[i+1][j+1][k+1]= E[i+1][j+1][k-6] + 10;  
S3:            E[i+1][j][k] = A[i-5][j-2][k+3] + C[i][j-2][k-2];  
S4:            D[i-3][j+1][k-2] = A[i-2][j-2][k] + 25  
        }  
    }  
}
```

- (1) Construct the dependency matrix/table, by showing the direction vectors.
- (2) Construct the dependency graph
- (3) Apply the code generation algorithm to extract parallelism from the loop nest. Show the step by step calculation of the algorithm execution.

Question-2:

[12 Points]

Consider the following loop nest:

```
for (int i=-10; i<=10; i++)  
{  
    A[i2+5i-13] = 10;  
    B[2i+1] = A[2i-3];  
}
```

Part-(a) Does the above loop have any dependency? Justify your answer.

Part-(b) Extract parallelism from the above loop. Show the parallelized loop nest.

Question-3:**[8 Points]**

Consider the following loop nest.

```
for ( int i =1; i <10; i ++ ) {
    m = A[i];
    n = A[i -1];
    for ( int j =1; j <20; j ++ ) {
        A[m] = A[i] + 10;
        A[n] = A[i -1] + 20;
    }
}
```

Part-(a): Does the above loop nest have dependency? Justify your answer.

Part-(b): Discuss how to extract parallelism in this case?

Question-4:**[25 Points]**

Consider the following loop nest.

```
for(int i =0; i <10; i ++ ) {
    for (int j =0; j <20; j ++ ) {
        for (k =0; k <30; k ++ ) {
            S1 : T=A[i ][j]+ C[i][j][k+2] + B[i][j][k -4]
            S2 : B[i][j][k] = T + C[i][j-1][k+2]
            S3 : C[i][j][k+2] = B[i][j-4][k-3] + d
        }
    }
}
```

Perform the following tasks in sequence

1. List down all RAW, WAR, and WAW dependencies in the loop nest.
2. Remove the dependencies associated with the scalar variable T without using scalar expansion.
3. Construct the dependency matrix after performing the above Step (2)
4. Perform the loop interchange optimization to extract fine-grained parallelism (at the innermost level) from the sequential program. Show the parallelized loop after performing the loop interchange.

Question-5:**[15 Points]**

Assume that you are given two graphs, $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$ both in the CSR format. Write a CUDA program (kernel is sufficient) to produce a new graph G, as defined below, in the CSR format.

$G(V, E) = G_1 \cup G_2$, where vertex set $V = V_1 \cup V_2$ and edge set $E = E_1 \cup E_2$

Question-6 (Short answer questions based on paper presentations): [15 Points]

1. List 3 applications of centrality metrics in graphs.
2. What are the optimizations techniques used in the parallel simulation of cellular automata on GPUs?
3. What are the benefits of sampling techniques in graph neural networks?
4. What are the issues of the CSR graphs and how does EFG solve them?
5. What are the different types of graph matching?
6. What are the challenges in maximal biclique enumeration on GPUs?
7. How does the multi-level graph partitioning work?
8. List a few limitations of the previous studies of subgraph matching.